Does Mamba earn its place?

Segmenting Metastatic Prostate Cancer in Whole-Body PET Scans with a Hybrid CNN-Mamba Network

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Supervised by Dr Jake Kendrick and Dr Mubashar Hassan

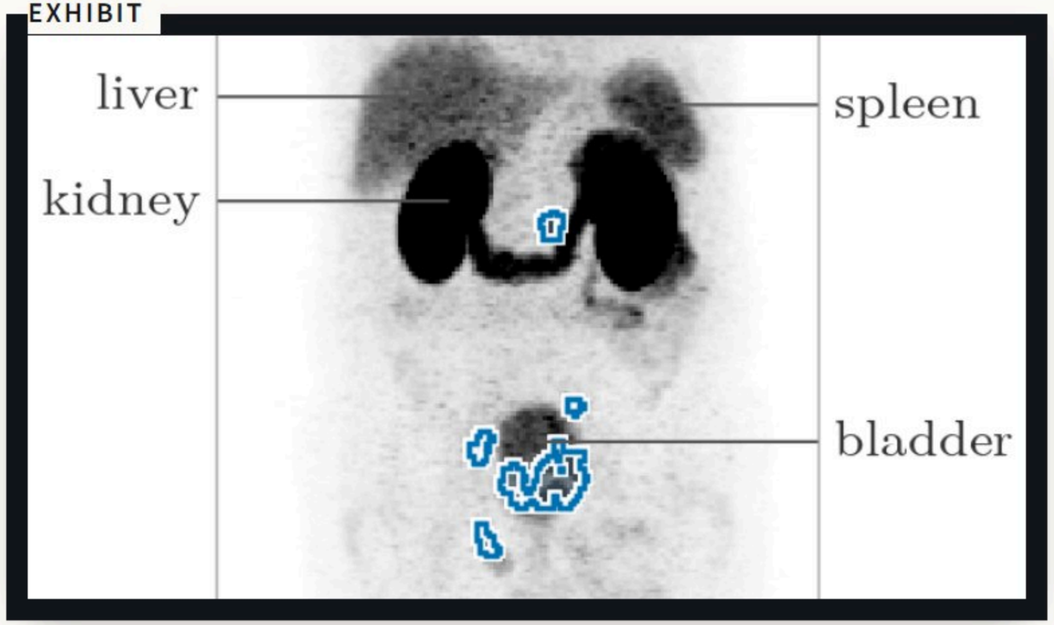
FOUR PREDICTIONS

- **H1** Detection
- **H2** Loss
- **H3** Efficiency
- **H4** Context

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Cancer hides among normal uptake, and experts outline it differently

EXHIBIT



autoPET-III PET projection, abdomen and pelvis. Dark is tracer uptake; lines name normal uptake; blue outlines are lesions.

TWO EXPERTS, THE SAME SCANS

33%

mean overlap (Dice) of their outlines

71.7%

of one expert's lesions found by the other

Second observer on 28 scans [1]

TERMS

Segmentation

outlining every lesion, voxel by voxel

[1] Kendrick et al., Eur J Nucl Med Mol Imaging 2022

Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

I built a hybrid network and tested it once on sealed scans

THE TALK IN ONE BREATH

PROBLEM	Outline every lesion in whole-body PSMA PET
IDEA	Mamba: cost grows linearly with length
NETWORK	Five stages, Mamba in the deepest three, plus organ training
DATA	697 scans, 130 sealed away
RESULT	**82% of lesions found at 86% precision**

TERMS

Precision

the share of predictions that hit a lesion

Convolutional networks read a scan one GPU-sized window at a time

nnU-Net [2]

a CNN that configures itself to each dataset

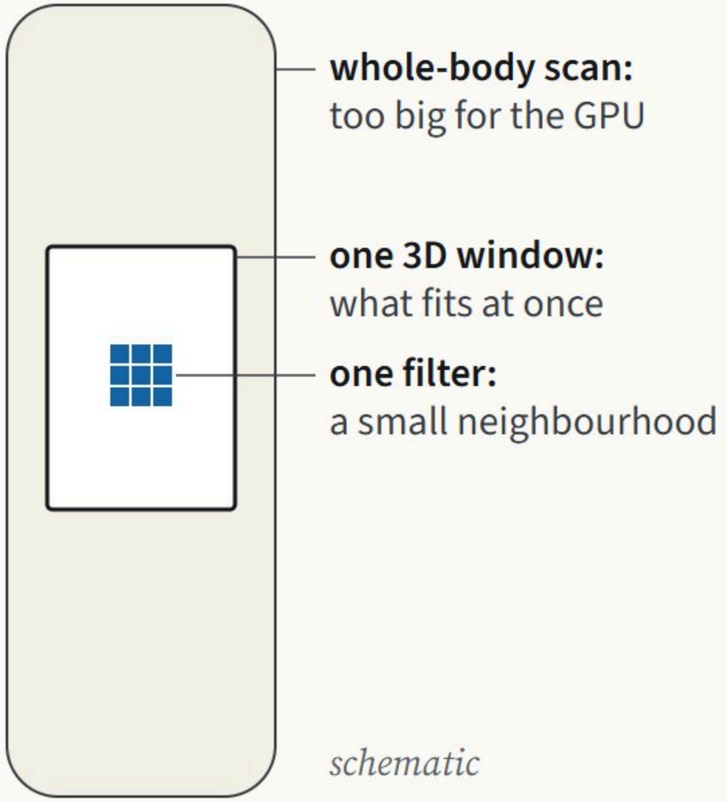
The 2022 nnU-Net cascade (Kendrick et al.) [1]

F1 79.9% precision 88.2% sensitivity 73.0%

its own 128 test scans; context only

PSMASegmentator MY COMPARATOR

a later nnU-Net, 1,035 training scans



schematic

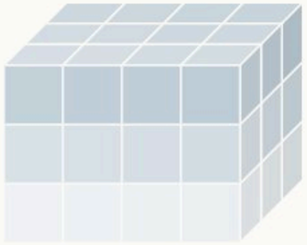
TERMS

CNN

a network of small filters that slide over the image

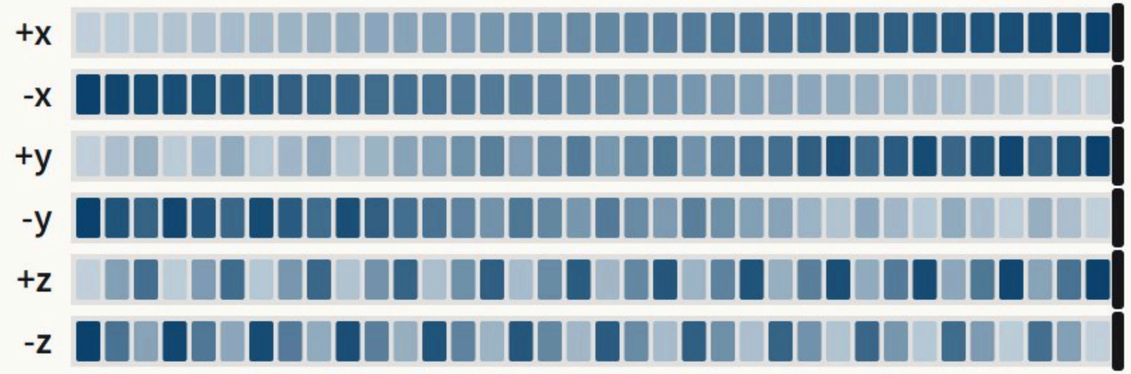
[2] Isensee et al., Nature Methods 2021 · [1] Kendrick et al., Eur J Nucl Med Mol Imaging 2022

Mamba reads a whole window's deep features at linear cost

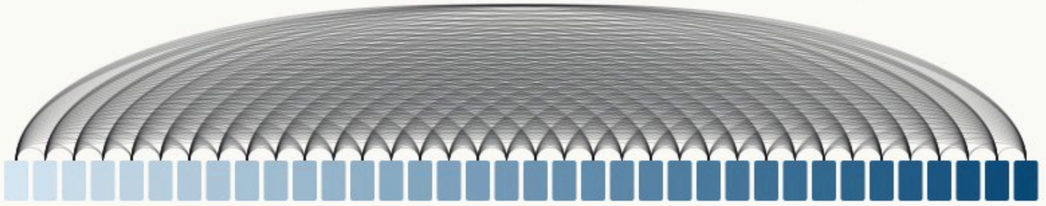


Deep features
(schematic)

Real tokens per order:
27,648, 3,456, 432



Mamba reads each order once: cost grows with length



Attention compares every pair: cost grows with length squared

> U-Mamba's Mamba layers had no effect against a copy without them [5]. A fair claim needs a CNN-only twin.

TERMS

Mamba

a sequence model whose cost grows linearly with length

[3] Gu and Dao, arXiv 2023 · [4] Ma et al., arXiv 2024 · [5] Isensee et al., MICCAI 2024

Four predictions turn one research question into tests

RESEARCH QUESTION

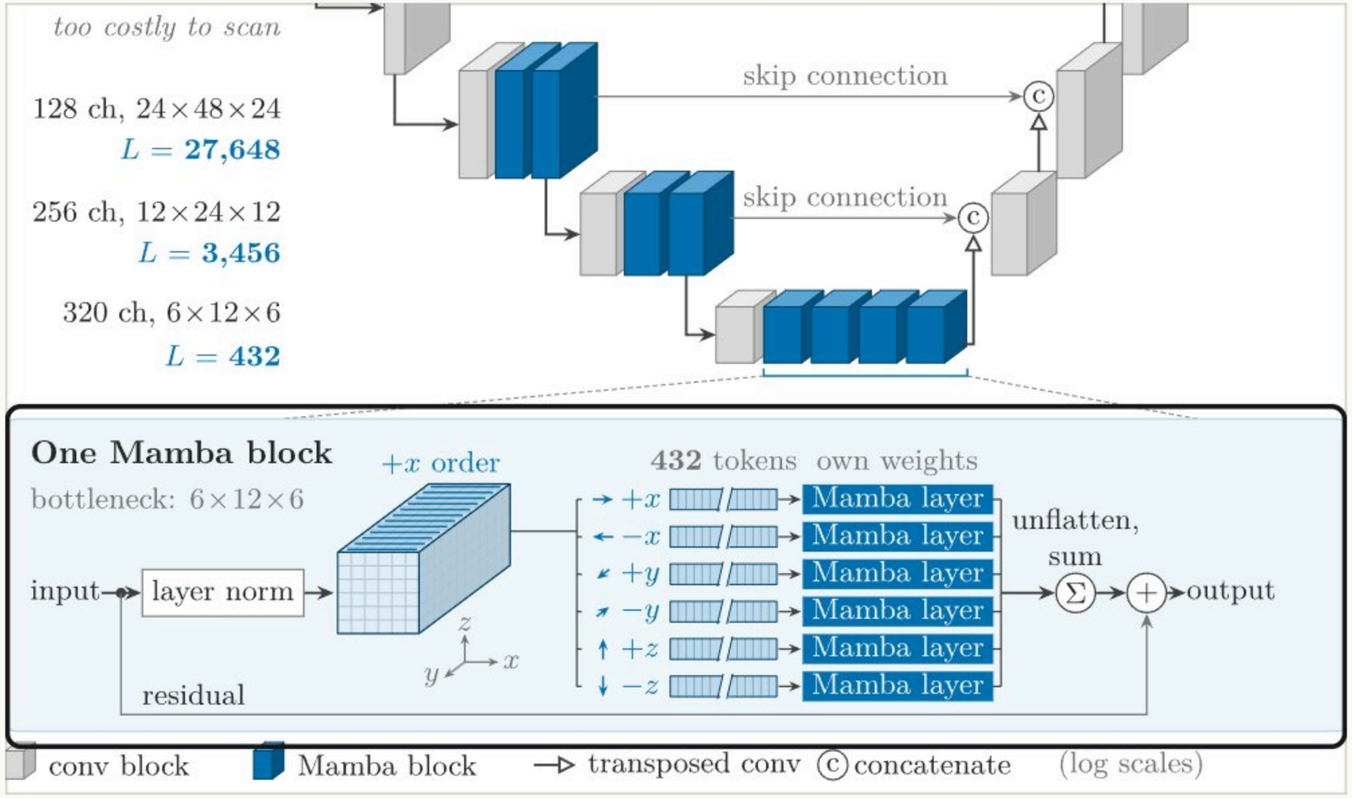
Does a hybrid CNN-Mamba network find lesions better than PSMASegmentator on scans neither model trained on, and do its Mamba blocks add anything over a CNN-only twin?

	PREDICTION	VERDICT
H1 Detection	The hybrid beats PSMASegmentator in pooled F1 on 100 scans neither model trained on.	◌ pending
H2 Loss	A loss weighted against misses lifts sensitivity and keeps precision high.	◌ pending
H3 Efficiency	At equal network passes, the hybrid is faster and uses less memory.	◌ pending
H4 Context	A bigger training window helps the hybrid more than its CNN-only twin.	◌ pending

HOW TO READ A VERDICT

- ◌ pending
- ◐ supported, as a trade
- ✕ not supported
- ← reversed on the primary set

My network keeps convolutions shallow and adds Mamba only deep



The network as drawn in the dissertation (Figure 3.3).

a Five stages, all convolutional

b Mamba only in the deepest three: 27,648 to 432 tokens

c Organ head (training only): false alarms on seven lesion-free scans, 21 to 7 [6]

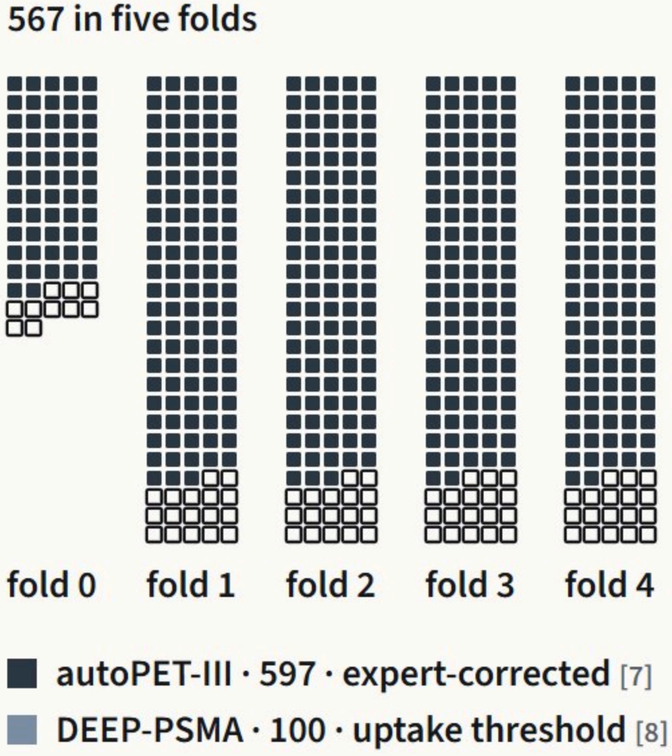
d One Mamba block: six scan orders, summed

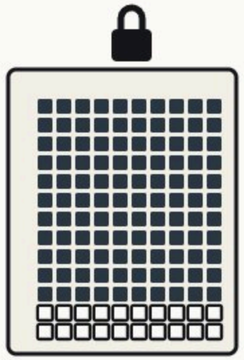
H1-H4

- H1
- H2
- H3
- H4

[6] Wasserthal et al., Radiol Artif Intell 2023

130 scans were sealed away and scored once, at the end





THE PROTOCOL

Sealed scans: never trained on

Five folds, five models, averaged into one ensemble

Pre-registered: rules fixed before testing

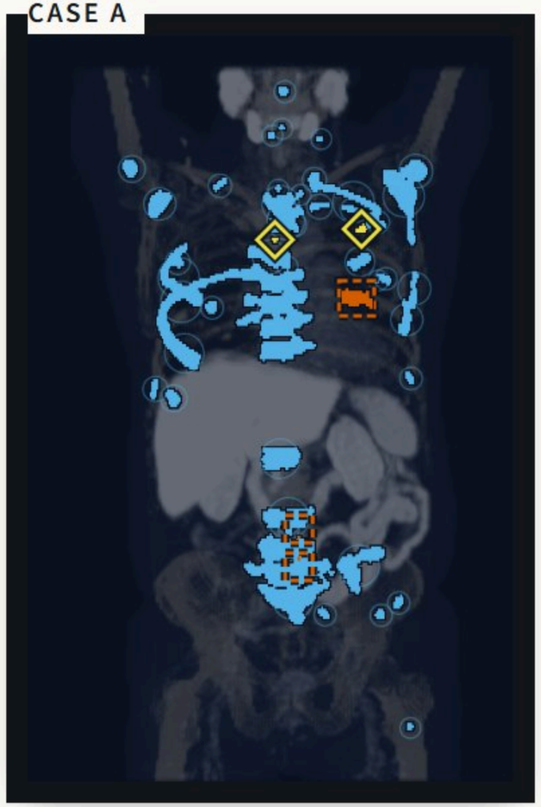
Run 4 never trained on DEEP-PSMA

H1-H4

- H1
- H2
- H3
- H4

[7] Jeblick et al., Sci Data 2026 · [8] Jackson et al., DEEP-PSMA design document 2025

On 130 sealed scans, the ensemble reached F1 84.3%



Case A: 49 of 52 found

CASE A WAS CHOSEN BY A RULE BEFORE VIEWING

found missed false alarm

ALL SEALED SCANS, LESIONS POOLED

84.3% F1

86.2%	82.4%	59.6%
precision	sensitivity	Dice

In-domain; scored once, as pre-registered

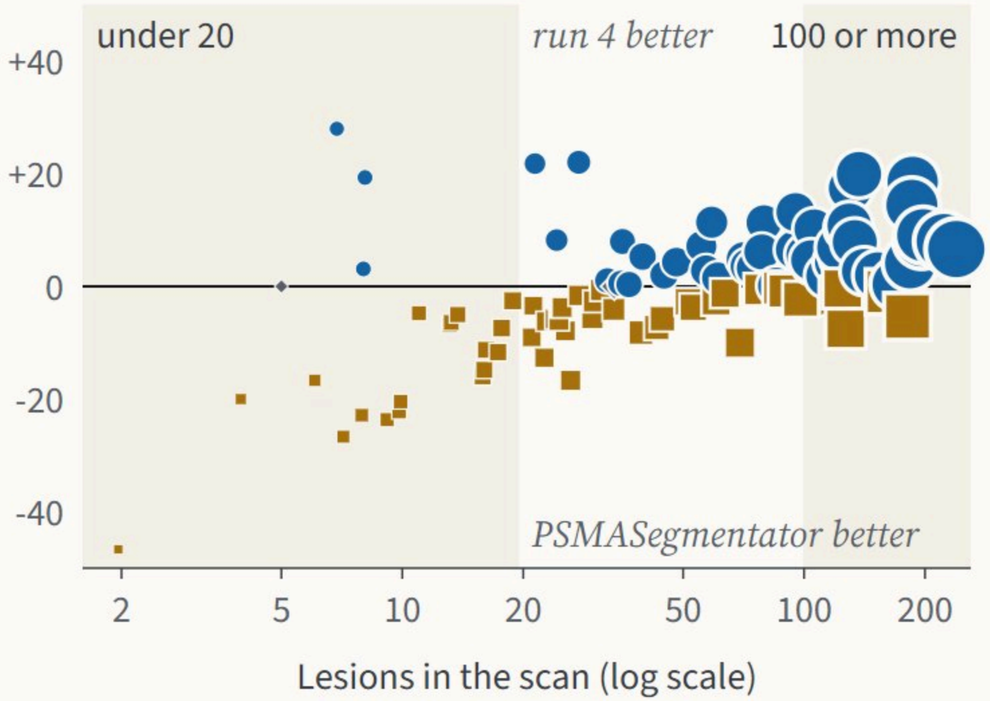
H1-H4

- H1
- H2
- H3
- H4

Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

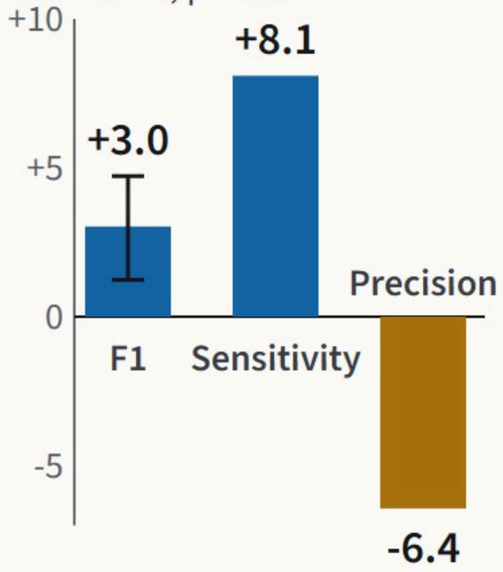
On unseen scans, run 4 found more lesions, with more false alarms

Each scan: run 4 minus PSMASegmentator, F1 points



Pooled over all lesions

difference, points



◐ **H1 supported, as a trade**: more lesions found, lower precision

FOUR PREDICTIONS

H1 Detection
◐ supported, as a trade

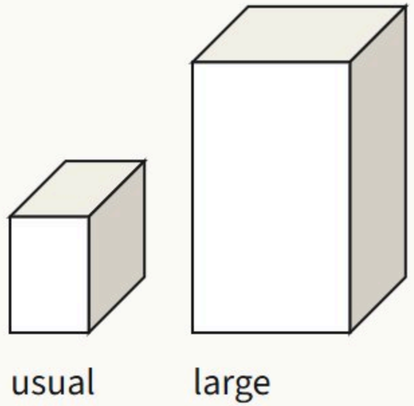
H2 Loss
◌ pending

H3 Efficiency
◌ pending

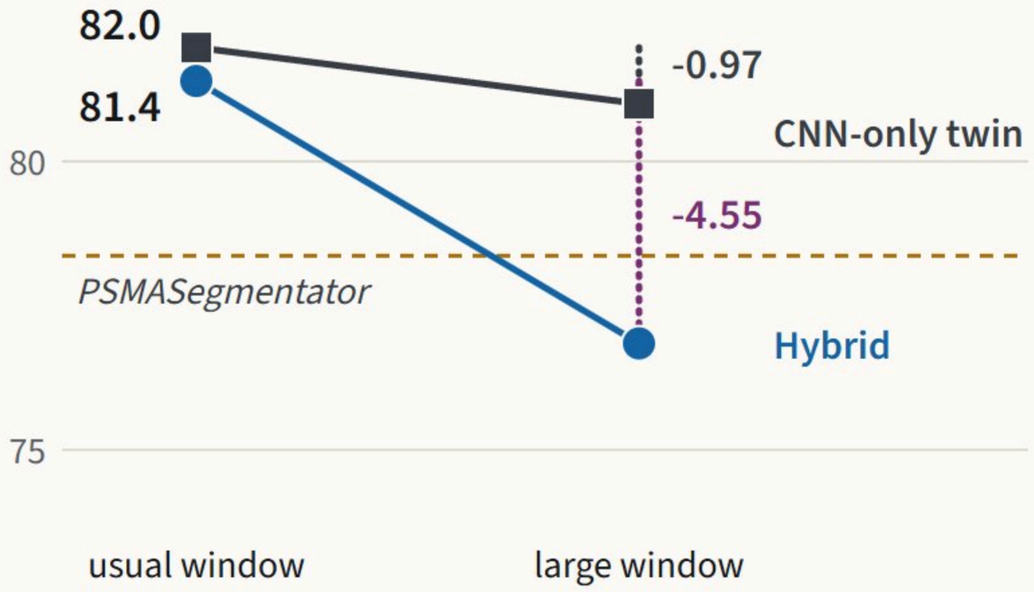
H4 Context
◌ pending

A bigger window hurt the hybrid more than its CNN-only twin

Training windows, to scale



F1 on the DEEP-PSMA scans



← **H4 reversed on the primary set**: interaction -3.57, the hybrid's drop minus the twin's; real under the pre-registered rule

FOUR PREDICTIONS

H1 Detection
◐ supported, as a trade

H2 Loss
◌ pending

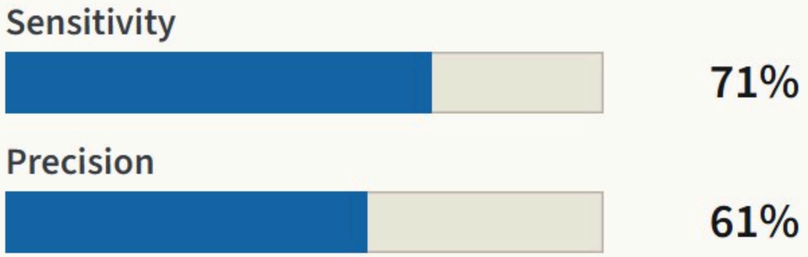
H3 Efficiency
◌ pending

H4 Context
← reversed on the primary set

The loss and speed predictions from my proposal were not supported

H2 · LOSS

Run 2: a loss weighted against misses [9]

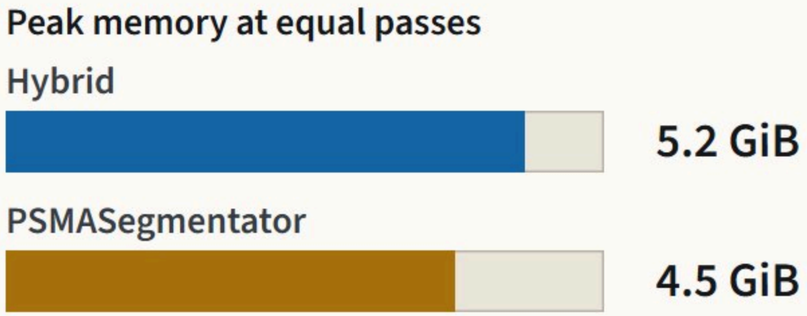


Partial test: run 2 alone, 57 validation scans, an older matcher

✕ **H2** not supported

H3 · EFFICIENCY, AT EQUAL PASSES

No faster in mean time per scan



The default-setup speed gap, and the abstract's, came from eight-pass test-time augmentation.

✕ **H3** not supported

FOUR PREDICTIONS

H1 Detection
◐ supported, as a trade

H2 Loss
✕ not supported

H3 Efficiency
✕ not supported

H4 Context
← reversed on the primary set

[9] Salehi et al., MLMI 2017

Run 4 beat PSMASegmentator in pooled F1, but Mamba takes no credit

Thank you

Questions?

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FOUR PREDICTIONS

H1 Detection
◐ supported, as a trade

H2 Loss
✕ not supported

H3 Efficiency
✕ not supported

H4 Context
← reversed on the primary set

WHAT IT MEANS

On DEEP-PSMA, run 4 found more lesions; the CNN-only twin did as well

LIMITATIONS

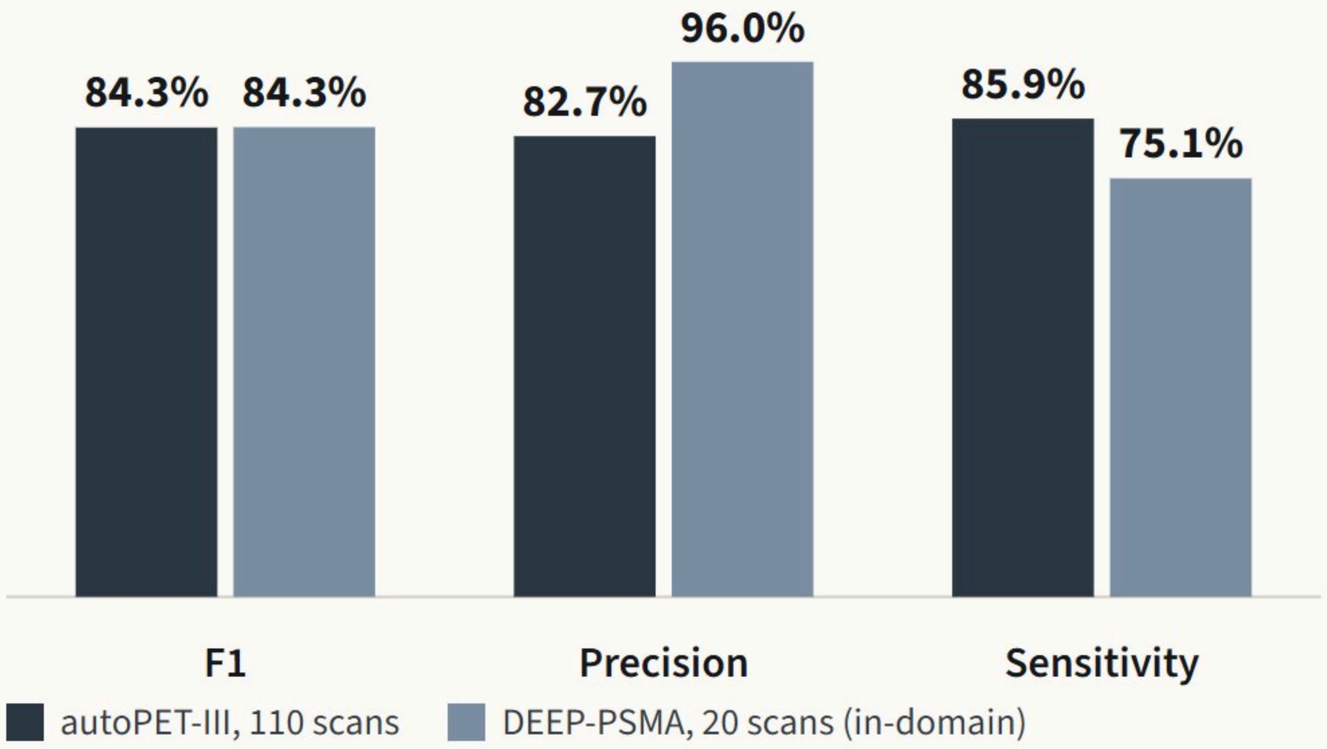
One run per cell; two label conventions; 8 lesion-free test scans

NEXT STEP

Retrain and score on the Sir Charles Gairdner Hospital cohort

Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

Both cohorts reach F1 84.3%, by different routes



- autoPET-III: expert-corrected outlines
- DEEP-PSMA: every voxel at SUV 3 or more, normal uptake painted out
- About 68 pieces per DEEP-PSMA scan

Cross-validation agreed with the sealed test

81.8%

Cross-validation, 567 scans pooled (fold sd 1.5)

84.3%

Sealed test, 130 scans (81.4 to 86.5)

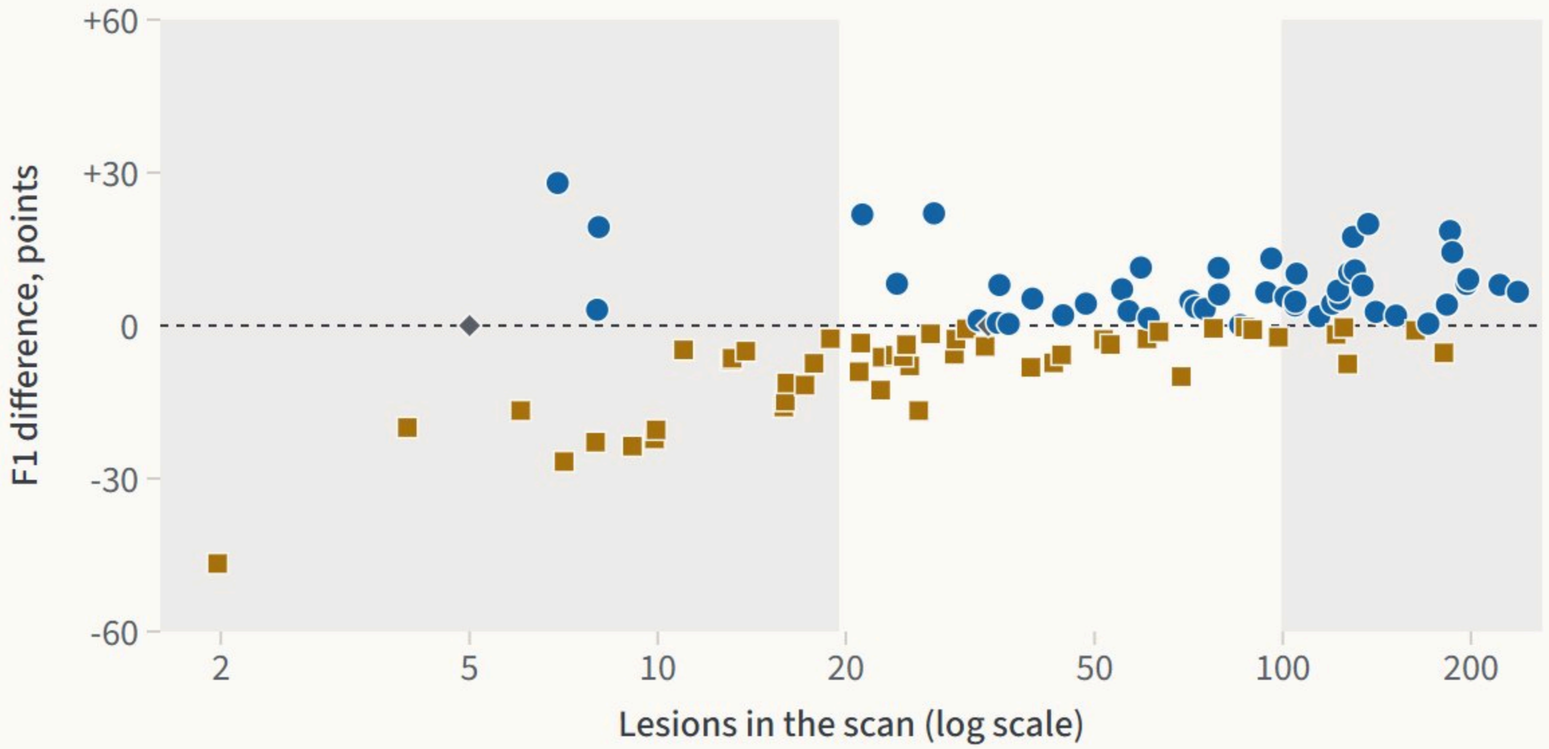
83.1

One fold model alone, plus or minus 0.6

Most recipe differences favour PSMASegmentator

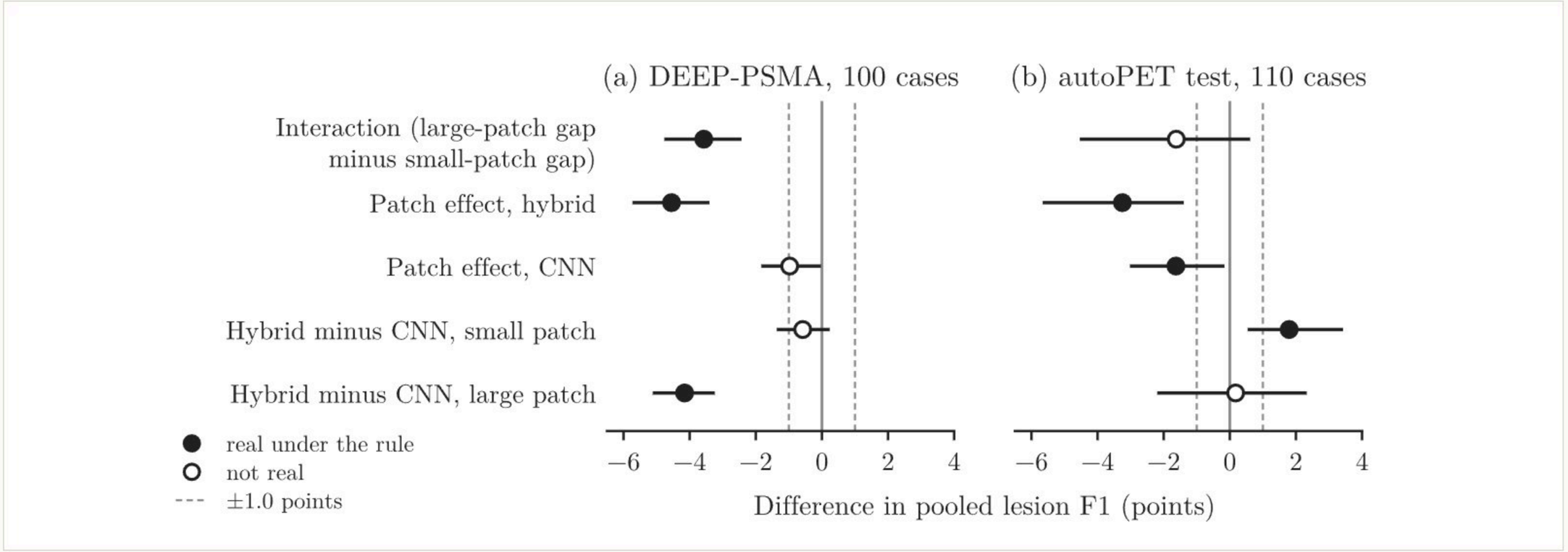
FACTOR	RUN 4 (MINE)	PSMASEGMENTATOR	FAVOURS
Training scans	430	1,035	PSMASegmentator
Models at inference	one	five averaged	PSMASegmentator
Test-time augmentation	off	eight mirrored passes	PSMASegmentator
Training window	1.77M voxels	4.72M voxels	PSMASegmentator
Parameters	39.7M	102.4M	PSMASegmentator
Optimiser	AdamW	SGD	unclear
Post-processing	none	off for this test	neither
Matching rule, grid, labels	same	same	neither

Scan by scan, there is no clear difference



Run 4 minus PSMASegmentator, per scan. Better on 48, worse on 50, tied on 2. Mean -0.6 (-2.8 to 1.6).

Every run 6 comparison, against the pre-registered rule



Real: interval excludes zero and size at least 1.0 point. One run per cell; large window at batch size 1; large hybrid selected at epoch 275.

The speed gap is test-time augmentation, not the architecture

CONFIGURATION	PASSES PER TILE	MEAN PER SCAN
Hybrid, as scored	5	18.1 s
PSMASegmentator, default	40	43.8 s
PSMASegmentator, augmentation off	5	17.2 s
Hybrid, augmentation on	40	131.2 s

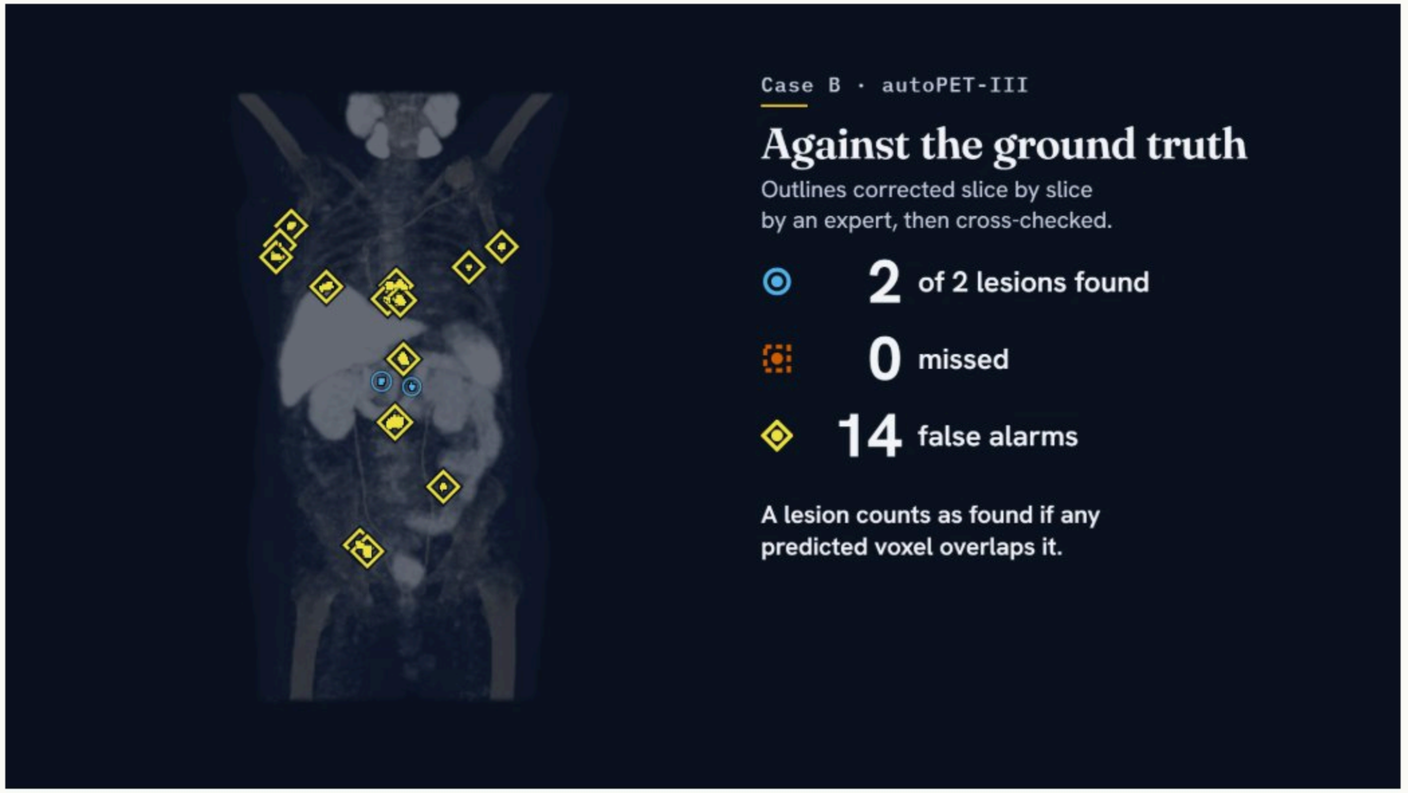
Development runs 1 to 4 fixed stability, resolution and false alarms

RUN	WHAT CHANGED	WHAT IT SHOWED
1	recall-weighted loss, no gradient clipping	loss spiked to 4,165; weights later lost
2	clipping, tiled validation (same loss)	precision about 61% (old matcher)
3	2 mm grid, Dice loss, bigger window	false alarms on all seven lesion-free scans
4	organ head	same predictions: F1 73.7 under the old matcher, 85.5 under any-overlap

Mamba sits only where the flattened sequence is short

STAGE	CHANNELS	TOKENS PER DIRECTION	BLOCK
1	32	1,769,472	convolution
2	64	221,184	convolution
3	128	27,648	convolution + 2 Mamba
4	256	3,456	convolution + 2 Mamba
5	320	432	convolution + 4 Mamba

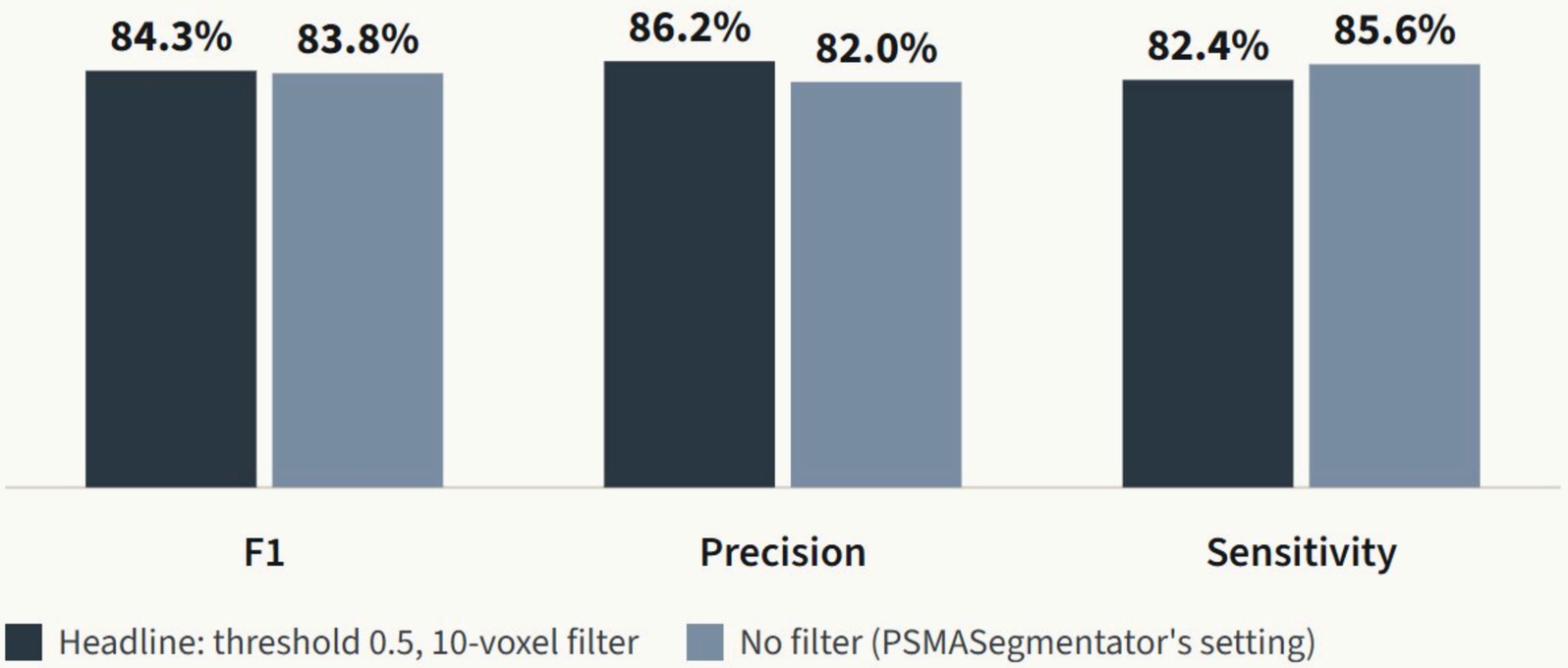
Case B: both lesions found, 14 false alarms



- 8 false alarms in bone, 6 in soft tissue
- None in the kidneys or bladder
- Not the worst scan: nine had no lesion found
- Chosen by rule before viewing

Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)

The threshold and size filter were fixed before the test



References

[1] J. Kendrick, R. J. Francis, G. M. Hassan, P. Rowshanfarzad, J. S. L. Ong, and M. A. Ebert, "Fully automatic prognostic biomarker extraction from metastatic prostate lesion segmentations in whole-body [68Ga]Ga-PSMA-11 PET/CT images," European Journal of Nuclear Medicine and Molecular Imaging, vol. 50, pp. 67-79, 2022.

[2] F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, "nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation," Nature Methods, vol. 18, no. 2, pp. 203-211, 2021.

[3] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," arXiv preprint arXiv:2312.00752, 2023.

[4] J. Ma, F. Li, and B. Wang, "U-Mamba: Enhancing long-range dependency for biomedical image segmentation," arXiv preprint arXiv:2401.04722, 2024.

[5] F. Isensee, T. Wald, C. Ulrich, M. Baumgartner, S. Roy, K. Maier-Hein, and P. F. Jaeger, "nnU-Net revisited: A call for rigorous validation in 3D medical image segmentation," in Medical Image Computing and Computer Assisted Intervention (MICCAI 2024), 2024.

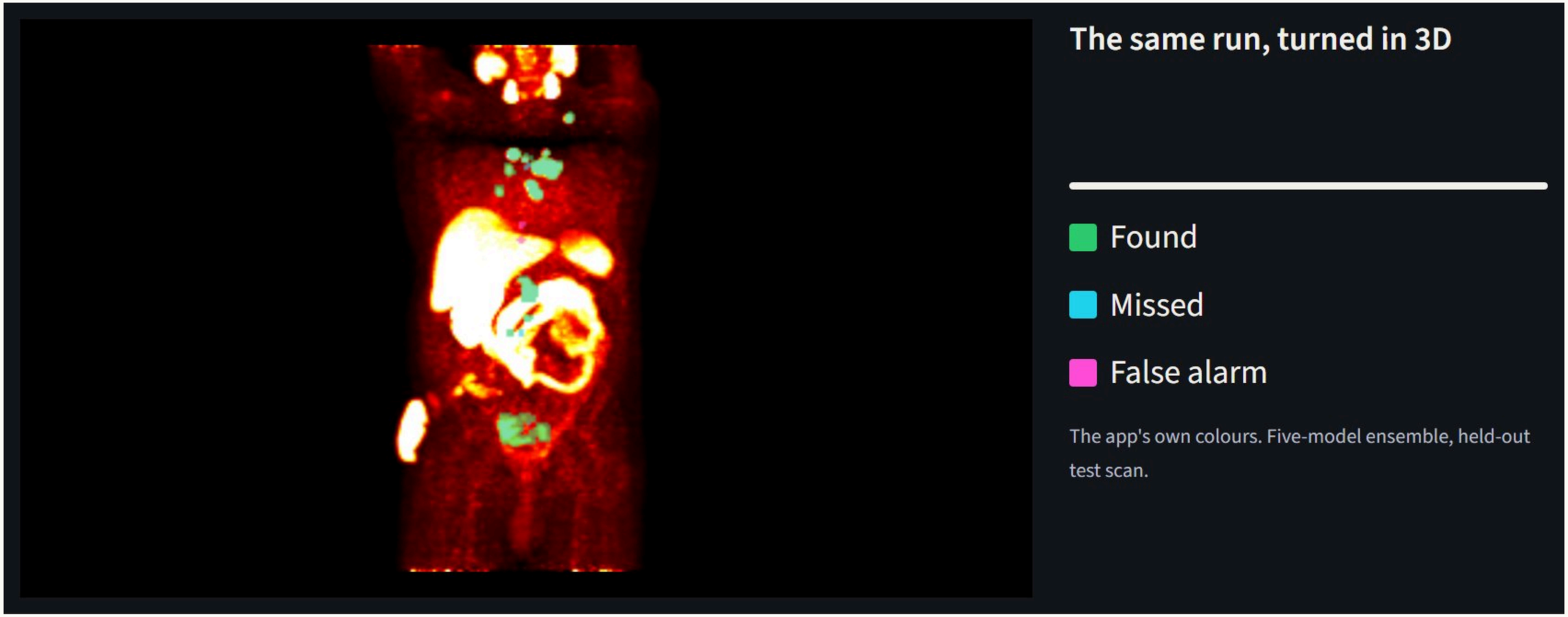
[6] J. Wasserthal, H.-C. Breit, M. T. Meyer, M. Pradella, D. Hinck, A. W. Sauter, T. Heye, D. T. Boll, J. Cyriac, S. Yang, M. Bach, and M. Segeroth, "TotalSegmentator: Robust segmentation of 104 anatomic structures in CT images," Radiology: Artificial Intelligence, vol. 5, no. 5, p. e230024, 2023.

[7] K. Jeblick, B. Schachtner, A. Mittermeier et al., "A whole-body PSMA-PET/CT dataset with manually annotated tumor lesions," Scientific Data, vol. 13, p. 1023, 2026.

[8] P. Jackson, L. McIntosh, J. Buteau, and M. Hofman, "Deep-learning evaluation for enhanced prognostics - prostate specific membrane antigen (DEEP-PSMA)," Challenge design document (Biomedical Image Analysis Challenges initiative), Zenodo, 2025, not peer-reviewed.

[9] S. S. M. Salehi, D. Erdogmus, and A. Gholipour, "Tversky loss function for image segmentation using 3D fully convolutional deep networks," in Machine Learning in Medical Imaging (MLMI), MICCAI Workshop, 2017.

The full recorded run in my app



Data: autoPET-III PSMA (Jeblick et al., Sci Data 2026, CC BY 4.0); DEEP-PSMA challenge (CC BY)